

EduSphere

Product Requirements Document (PRD)

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1. PROJECT OVERVIEW

The company aims to build a modern e-learning platform that allows instructors to create and sell online courses, and enables students to learn, track their progress, and obtain completion certificates.

⚠ Note: *Scope (Phase 1): Web platform only.*

2. USER MANAGEMENT

2.1 Registration

A user can create a new account using the following fields:

- First Name
- Last Name
- Username
- Email
- Password

Constraints:

- Username must be unique.
- Email must be unique.

Post-registration:

- A verification email is sent to the user's email address.
- The user cannot access the platform before verifying their email.

2.2 Authentication

A user can log in using:

- Username or Email
- Password

The system must support:

- Refresh Tokens
- Logout from the current device
- Logout from all devices

2.3 Roles

The system supports the following roles:

- Student
- Instructor
- Admin

⚠ Note: *A user can hold only one role at a time.*

2.4 Profile Management

A user can:

- Edit their personal information
- Upload a profile picture
- Write a bio / self-introduction
- Set their country
- Set their specializations or educational interests

3. COURSE MANAGEMENT

3.1 Create Course

An instructor can create a new course containing the following information:

- Title
- Description
- Thumbnail Image
- Price
- Language
- Level
- Requirements
- Learning Objectives

 **Note:** A newly created course defaults to "Draft" status.

3.2 Course Status

A course can be in one of the following statuses:

- Draft
- Pending Review
- Published
- Rejected
- Archived

 **Note:** Students cannot see unpublished courses.

3.3 Categories

A course can be classified under one or more categories.

Examples:

- Programming
- Design
- Business
- Marketing

3.4 Sections

A course consists of multiple Sections. An instructor can:

- Add a Section
- Edit a Section
- Delete a Section
- Reorder Sections

3.5 Lessons

Each Section contains multiple Lessons. A lesson can be of the following types:

- Video
- Article
- PDF
- Downloadable Resource

 **Note:** Lessons can be reordered within a Section.

4. COURSE ENROLLMENT

4.1 Purchase Course

- A student can purchase a paid course.
- Upon purchase, a new Order is created.

4.2 Enrollment

After a successful payment:

- The student is enrolled in the course.
- The student gains access to all course content.
- The enrollment date is saved.

 **Note:** A student cannot be enrolled in the same course more than once.

5. PAYMENT MANAGEMENT

5.1 Payment Provider

The system uses Paymob as the primary payment gateway.

5.2 Orders

When a purchase process is initiated, an Order is created containing:

- Order Number

- Total Amount
- Status

5.3 Payment Status

A payment can have one of the following statuses:

- Pending
- Paid
- Failed
- Cancelled
- Refunded

5.4 Payment Callback

- Paymob sends a callback after the payment process completes.

On successful payment:

- The payment status is updated.
- An Enrollment record is automatically created.

6. LEARNING PROGRESS

6.1 Lesson Completion

- A student can mark a lesson as completed.

6.2 Progress Tracking

The system must save:

- Number of completed lessons
- Completion percentage
- Last lesson viewed

7. QUIZZES

7.1 Create Quiz

An instructor can create a Quiz within a course. A Quiz contains:

- Title
- Duration
- Passing Score

7.2 Question Types

A Quiz can contain the following question types:

Multiple Choice Question:

- Question Text
- Multiple Options
- Correct Option

True / False Question:

- Question Text
- Correct Answer

7.3 Quiz Attempts

- A student can attempt a quiz.

The system must save:

- Start Time
- End Time
- Score
- Result
- A student can retry the quiz if the instructor allows it.

8. CERTIFICATES

8.1 Course Completion Criteria

A student is considered to have completed a course when they have:

- Completed all required lessons.
- Passed all required quizzes.

8.2 Certificate Generation

Upon course completion, a certificate is generated containing:

- Certificate Number
 - Student Name
 - Course Name
 - Issue Date
 - The student can download the certificate.
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9. REVIEWS & RATINGS

9.1 Ratings

A student can rate a course on a scale of:

- 1 Star
- 2 Stars
- 3 Stars
- 4 Stars
- 5 Stars

 **Note:** A student cannot rate the same course more than once.

9.2 Reviews

- A student can write a text review for a course.
 - The student can edit or delete their review.
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10. WISHLIST

- A student can add a course to their wishlist.
 - A student can remove a course from their wishlist at any time.
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11. NOTIFICATIONS

Notifications must be created for the following events:

- A new course is published
- A course is purchased
- A quiz is passed
- A certificate is obtained
- A new review is added to a course

A user can:

- View their notifications
 - Mark notifications as read
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12. SEARCH

The platform must support searching for:

- Courses
- Instructors

- Categories
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13. ADMINISTRATION

13.1 User Moderation

An admin can:

- Disable a user account
- Re-activate a user account

13.2 Course Moderation

An admin can:

- Review courses
- Approve a course
- Reject a course
- Archive a course

13.3 Payment Monitoring

An admin can:

- Monitor payment transactions
 - Review orders
 - Review refund operations
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14. AUDIT REQUIREMENTS

The following timestamps must be stored for all major data entities:

- Created At
 - Updated At
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15. NON-FUNCTIONAL REQUIREMENTS

15.1 Security

- JWT Authentication
- Role-Based Authorization
- Email Verification
- Secure Paymob Integration

15.2 Performance

The system must support:

- 10,000 concurrent users
- 5,000 courses
- 100,000 enrollment operations

15.3 Scalability

The system must allow adding additional payment providers in the future without requiring core system changes (open/closed design principle).

15.4 Real-Time

Notifications must operate in real-time.

16. DDD ANALYSIS INSTRUCTIONS

 **Note:** Do NOT open SQL or EF Core yet.

Work in the following order:

1. Identify the Bounded Contexts.
2. Extract the Aggregates.
3. Determine the Aggregate Roots.
4. Extract the Entities.
5. Extract the Value Objects.
6. Draw the Domain Model.
7. Draw the Context Map.
8. Write the Domain Events.
9. Only then, create the ERD.